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A Associative Memories with Random Vectors

In this section, we provide basic properties of associative memories based on outer products of random Gaussian embeddings, as described in Section 4.

We consider embeddings $u_k \in \mathbb{R}^d$ with i.i.d. Gaussian $\mathcal{N}(0, \frac{1}{d})$ entries.

We recall a few facts:

- (*Norm*) We have $u_i^\top u_i \approx 1$. This is standard from the concentration of random vectors in high dimension (see, *e.g.*, [54, Theorem 3.1.1]).
- (*Near-orthogonality*) For $i \neq j$, we have $u_i^\top u_j = O(1/\sqrt{d})$. To see this, denoting $u_i = d^{-1/2}(\tilde{u}_{ik})_k$, where $\tilde{u}_{ik}$ are the unnormalized entries of u_i, note that we have

$$\sqrt{d} u_i^\top u_j = \frac{1}{\sqrt{d}} \sum_{k=1}^{d} \tilde{u}_{ik} \tilde{u}_{jk} \to \mathcal{N}(0,1),$$

  by the central limit theorem, since for each k, the quantities $\tilde{u}_{ik}\tilde{u}_{jk}$ are zero-mean, unit-variance, i.i.d. random variables.
- (*Remapping: norm*) If W is a Gaussian random matrix with i.i.d. $\mathcal{N}(0, \frac{1}{d})$ entries, then for any fixed x we have $\|Wx\| \approx \|x\|$. This follows from Johnson-Lindenstrauss (see, *e.g.*, [54, Lemma 5.3.2 and Exercise 5.3.3]). In particular, if x is a normalized Gaussian embedding as above, then $\|Wx\| \approx 1$.
- (*Remapping: near-orthogonality*) Consider a random vector $x = \frac{1}{\sqrt{d}}\tilde{x}$ and a random matrix $W = \frac{1}{\sqrt{d}}\tilde{W}$, where the entries of $\tilde{x}$ and $\tilde{W}$ are i.i.d. $\mathcal{N}(0,1)$. Then x and Wx are nearly orthogonal. To see this, note that $\mathbb{E}[x^\top W x] = \mathbb{E}[x^\top \mathbb{E}[W] x] = 0$, and the variance is

$$\begin{aligned}\mathbb{E}(x^\top W x)^2 &= \mathbb{E}\sum_{i,j} x_i^2 W_{ij}^2 x_j^2 = \frac{1}{d^3}\,\mathbb{E}\sum_i \tilde{x}_i^4 \tilde{W}_{ii}^2 + \mathbb{E}\sum_{i\neq j} \tilde{x}_i^2 \tilde{x}_j^2 \tilde{W}_{ij}^2 \\ &= \frac{1}{d^3}\left(dM_4M_2 + \frac{d(d-1)}{2}M_2^3\right) = O\left(\frac{1}{d}\right),\end{aligned}$$

  where M_2 and M_4 denote the second and fourth moments of the standard Gaussian, respectively. Then, Chebyshev's inequality implies that $|x^\top W x| = O(1/\sqrt{d})$ with high probability.

Ensuring appropriate memory lookups then requires such properties to hold for many embeddings and pairs of embeddings, with errors that are small enough to ensure correct associations. This may be achieved with careful union bounds or more powerful concentration results. We do not attempt to do this in a precise manner in this paper, and will generally assume d large enough to satisfy the desired associative memory behaviors, noting that a precise analysis is an important direction for future work (see [8] for follow-up work in this direction).

B Theoretical Insights on Gradient Dynamics

In this section, we provide additional details on the theoretical insights from Section 6, including details on the illustrative example (Section B.1), derivations of gradients w.r.t. key-query matrices at initialization (Section B.2), as well as a study of how the induction head mechanism may develop in a simplified setup, using a sequence of single layer-wise gradient steps in a top-down manner (Section B.3).

B.1 Details on illustrative example

Consider the example discussed in Section 6: we would like to predict y from $x = w_E(y) + p_t$, where p_t is a positional embedding at a random position $t \in [T]$, which we would like to ignore. Further assume that y is uniformly distributed ($p(y = k) = 1/N$) and consider the matrix obtained after one population gradient step with step-size η starting from an initialization $W_0 = 0$ (so

that $\hat{p}_{W_0}(k|x) = 1/N$):

$$W_1 = \frac{\eta}{N} \sum_{k=1}^{N} w_U(k)(\mu_k - \bar{\mu})^\top, \tag{11}$$

with $\bar{\mu} = \mathbb{E}[x]$.

Note that we have $\mu_k = w_E(k) + \frac{1}{T}\sum_t p_t$ and $\bar{\mu} = \frac{1}{N}\sum_k w_E(k) + \frac{1}{T}\sum_t p_t$, so that (11) becomes

$$W_1 = \frac{\eta}{N} \sum_{k=1}^{N} w_U(k)(w_E(k) - \bar{w}_E)^\top, \tag{12}$$

with $\bar{w}_E := \frac{1}{N}\sum_{k=1}^{N} w_E(k)$. When d is large enough to ensure near-orthonormal embeddings, we have for any y and t,

$$W_1(w_E(y) + p_t) \approx \frac{\eta}{N} w_U(y) + O\left(\frac{1}{N^2}\right).$$

This implies

$$w_U(k)^\top W_1(w_E(y) + p_t) \approx \frac{\eta}{N}\, \mathbb{1}\{k = y\} + O\left(\frac{1}{N^2}\right),$$

as claimed in the main text. The classifier $\hat{y} = \arg\max_k w_U(k)^\top W_1(w_E(y)+p_t)$ then has essentially perfect accuracy, and has learned to ignore the spurious positional embeddings, which are simply exogenous noise.

B.2 Gradients on key-query matrices at initialization

We now derive expressions for population gradients of the attention key-query matrices at zero initialization, noting that random initialization behaves similarly to zero initialization. Although the optimization problems involving these matrices are non-convex, these gradients at initialization lead to associative memories, similar to Lemma 2. When output matrices of the previous layer already encode the desired associations, these gradients can lead to associative memories that focus the attention on the correct key-value pairs.

We begin with the following lemma, which gives the gradient of the loss w.r.t. $W = W_K^2$ at zero initialization. For simplicity, we drop the $d^{-1/2}$ factor from the softmax, which only changes gradients by a multiplicative factor, and thus does not change its form.

Lemma 4 (Gradient of second attention layer)**.** *Consider the following loss for predicting the next token y from an attention layer with inputs $X = [x_1, \ldots, x_T]$, and value-output matrix $\Phi_2 := W_O^2 W_V^2$:*

$$L(W) = \mathbb{E}_{(X,y)}[\ell(y, \xi(X))], \qquad \xi(X) = W_U \Phi_2 X \sigma(X^\top W x_T), \tag{13}$$

with ℓ the cross-entropy loss and $\sigma(u)_t = \frac{e^{u_t}}{\sum_s e^{u_s}}$ for $u \in \mathbb{R}^T$ is the softmax.

The gradient at $W = 0$ is given by

$$\begin{aligned}
\nabla_W L(W)\big|_{W=0} &= \sum_{k=1}^{N} \mathbb{E}_{(X,y)}\left[(\hat{p}_W(k|X) - \mathbb{1}\{y{=}k\}) \frac{1}{T}\sum_{t=1}^{T} w_U(k)^\top \Phi_2 x_t \cdot (x_t - \bar{x}_{1:T}) x_T^\top\right] \\
&= \frac{1}{T}\sum_{k=1}^{N}\sum_{t=1}^{T} \mathbb{E}_X[\hat{p}_W(k|X) w_U(k)^\top \Phi_2 x_t \cdot (x_t - \bar{x}_{1:T}) x_T^\top] \\
&\quad - \frac{1}{T}\sum_{k=1}^{N}\sum_{t=1}^{T} p(y=k)\, \mathbb{E}_X[w_U(k)^\top \Phi_2 x_t \cdot (x_t - \bar{x}_{1:T}) x_T^\top \mid y = k]
\end{aligned}$$

with $\bar{x}_{1:T} = \frac{1}{T}\sum_{t=1}^{T} x_t$.

Now we consider the gradient w.r.t. $W = W_K^1$ at zero initialization, and consier a simplification of the second layer attention to its linearization around $W_K^2 = 0$. We will see that this still provides first-order information that is sufficient for W_K^1 to be learned.